



LAN743x and LAN887x QNX 8 io-sock Driver User Guide

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1 Overview

1.1 Introduction

This document comprises of release, build and installation instructions of QNX 8 io-sock driver for LAN743x and LAN887x devices.

LAN7430 device is a PCIe Gigabit Ethernet controller with integrated MAC and Phy. The LAN743x QNX driver supports the LAN7430 device. LAN7431 is a PCIe Gigabit controller with MAC. An external phy need to be connected via RGMII interface. In this driver package, LAN887x phy driver is provided for use with LAN7431 device.

This document assumes readers are familiar with the QNX 8 driver environment, development system and Momentics IDE.

1.2 References

1. LAN7430/LAN7431 Data sheet.
2. LAN8870/LAN8871/LAN8872 Data sheet

1.3 Terms and Abbreviations

Table 1 Abbreviations

IDE	Integrated Development Environment
MAC	Media Access Control
Phy	Physical Layer Device
SDP	Software Development Platform
SQI	Signal Quality Indicator

2 Driver package

Extract the driver .zip package. The driver package contains folders below.

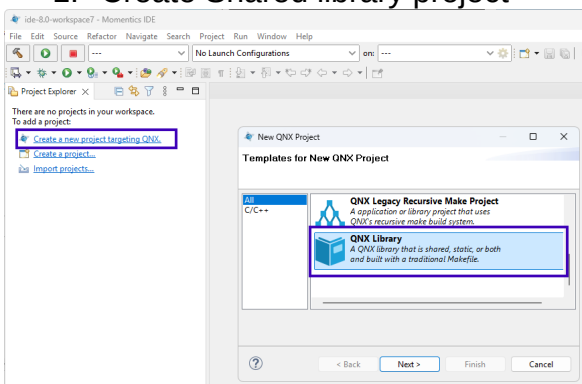
lan743x	- LAN743x QNX 8 io-sock driver files
phys/lan887x	- LAN887x Phy QNX 8 io-sock driver files
docs	- This document and Release Notes

3 QNX 8 Momentics

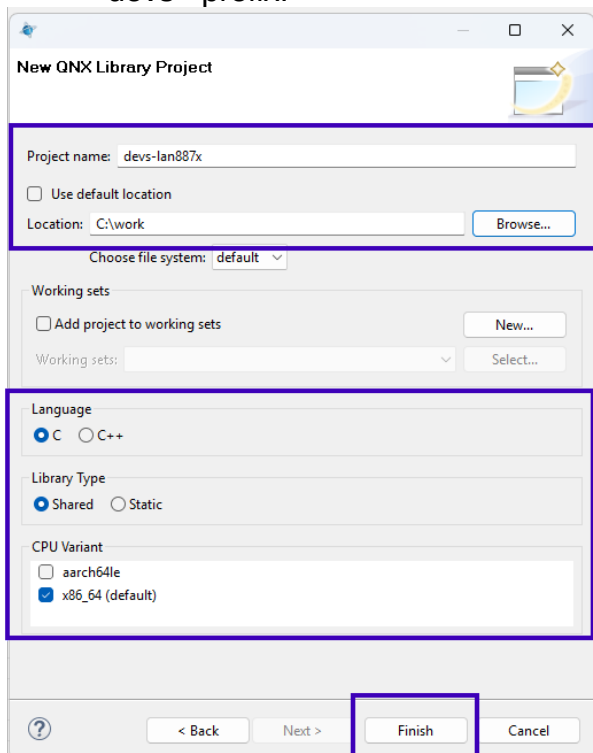
The LAN743x and LAN887x drivers can be built in QNX Momentics IDE. This section assumes that the QNX 8 SDP and Momentics IDE installed.

3.1 LAN887x driver project creation

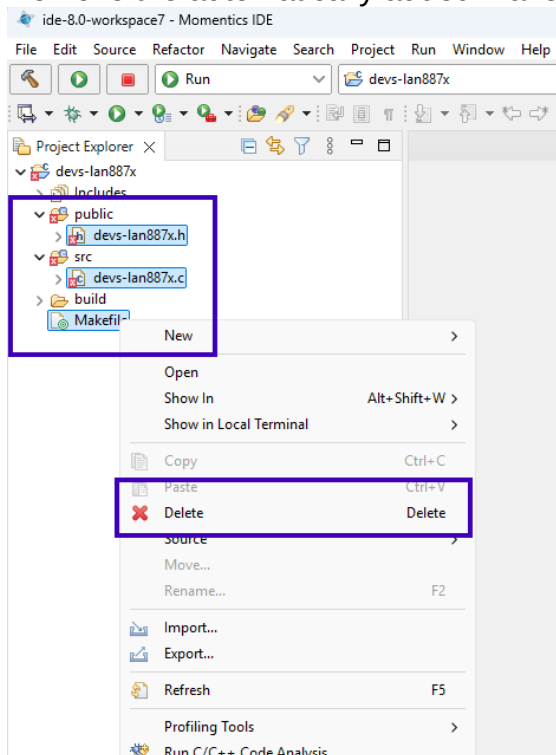
1. Create Shared library project



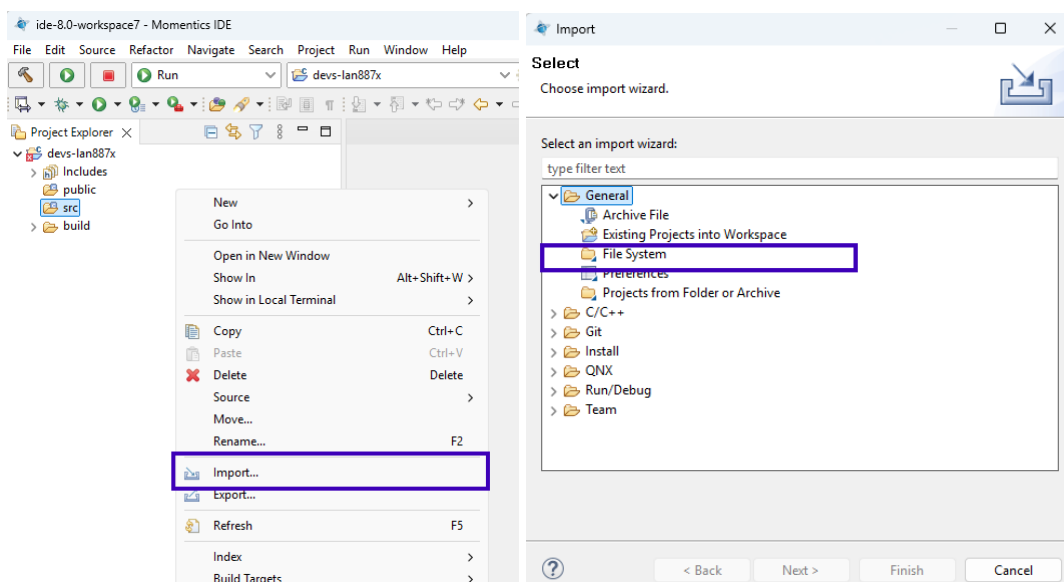
2. Provide name *devs-lan887x* and select the architecture. Note: The name must have 'devs-' prefix.

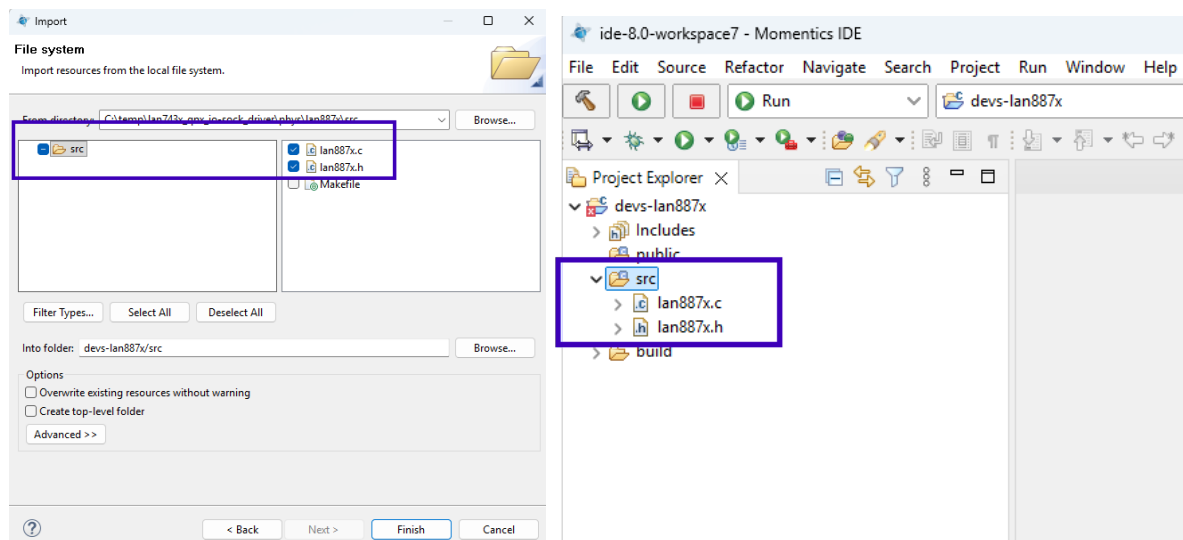


3. Remove the automatically added Makefile, source and header files.

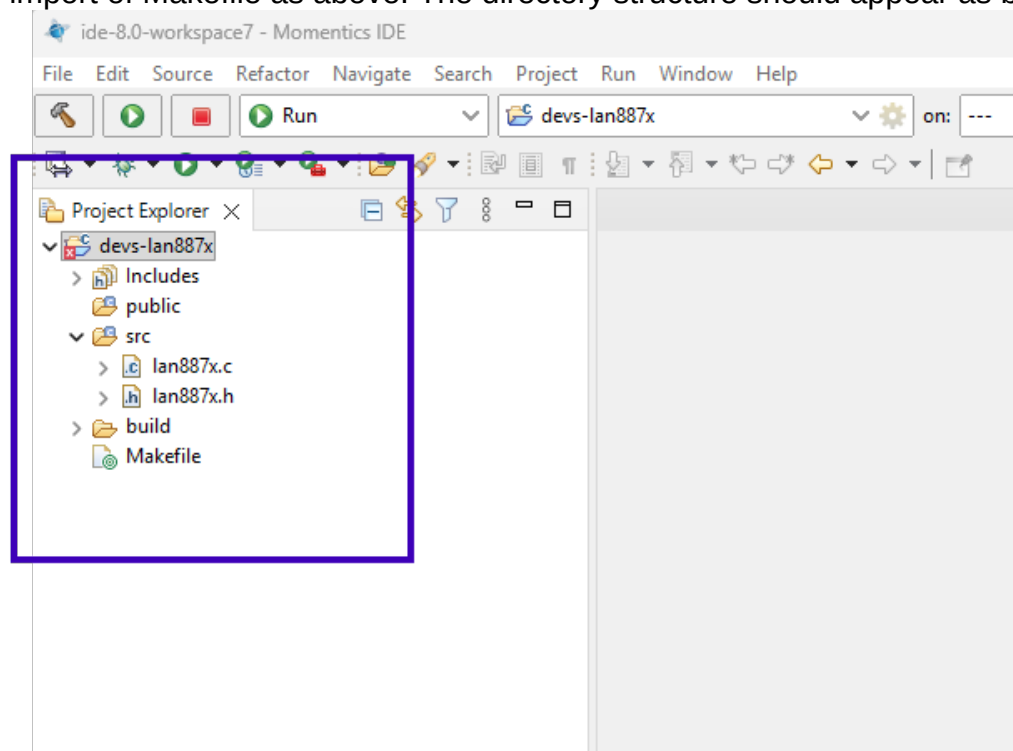


4. Add source and header files from the release package. Right click on `src` and do import.



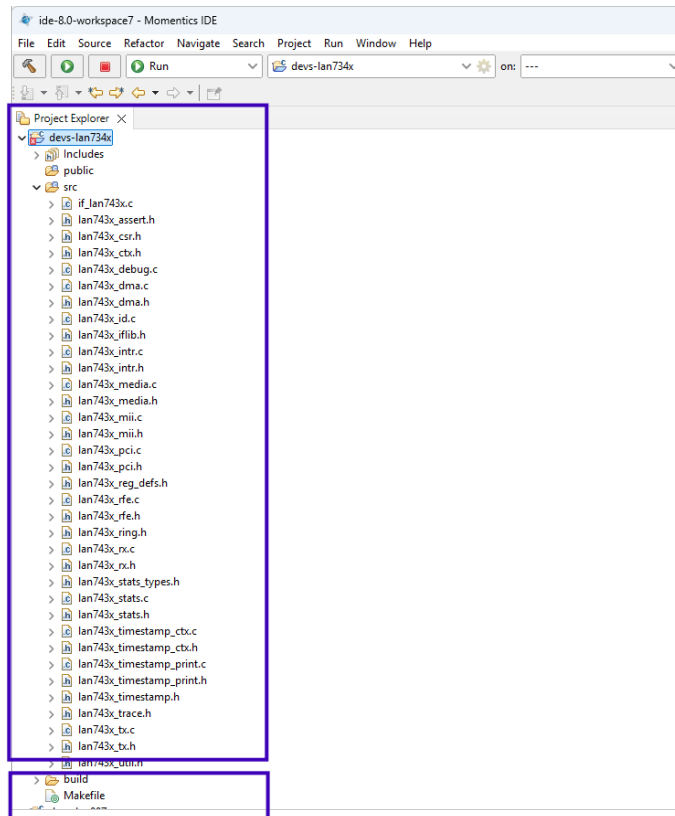


5. Add Makefile from the release package. Highlight the *devs-lan887x* folder and do import of Makefile as above. The directory structure should appear as below.



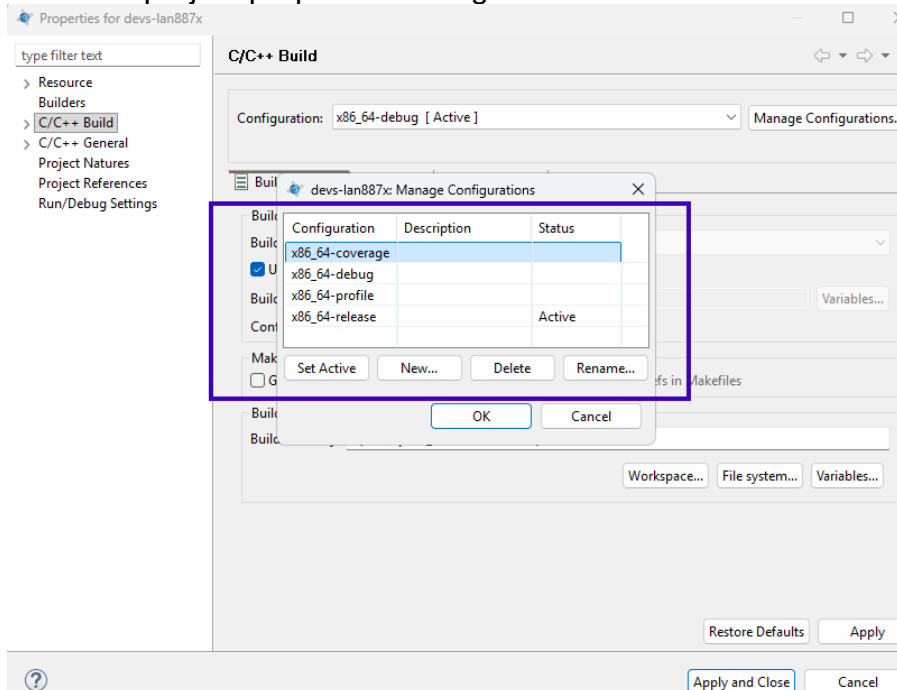
3.2 LAN743x driver project creation

1. Create Shared library project similar steps mentioned in the previous section.
2. Provide name *devs-lan743x* and select the architecture. Note: The name must have 'devs-' prefix.
Note: The interface name appears as *lan743x0*. If the smaller name is desired, provide the name ex: *devs-lan*. The interface name shown in ifconfig appears as *lan0*. In this case, the ARTIFACT variable in the Makefile also must be modified.
3. Remove the automatically added Makefile, source and header files.
4. Add Makefile, source and header files from the release package.



3.3 Release build and debug build settings

From the project properties debug or release build can be selected.



3.4 Driver build

The driver .so file will be built under debug or release folder depending on the IDE setting.

4 Driver Installation

4.1 Remote target connection

The driver .so files can be transferred to target system either ssh or via *Target File System Navigator* panel with the IDE. Transfer the files to /tmp folder of the target system.

4.2 Driver mount

Once the driver is on the target system it must be attached to the network stack before the interface can be used. This is done using the mount command.

```
# mount -T io-sock /tmp/devs-lan887x.so
# mount -T io-sock /tmp/devs-lan743x.so
```

This will create a network interface for LAN743x found on the PCI bus. *ifconfig* command can be used to check the interface.

```
# ifconfig
```

```
lan743x0: flags=8822<BROADCAST,SIMPLEX,MULTICAST> metric 0 mtu 1500
options=800103<RXCSUM, TXCSUM, TS04>
ether 00:80:0f:74:31:00
media: Ethernet 1000BASE-T1 <full-duplex, master>
nd6 options=21<PERFORMNUD, AUTO_LINKLOCAL>
```

4.3 MAC Address and IP address

The LAN743x driver reads the EEPROM of the device and sets the MAC address. Note: Changing MAC address through *ifconfig* is not supported in this release.

The IP address can be assigned using *ifconfig* command.

```
# ifconfig lan743x0 192.150.0.100/24
# ifconfig lan743x0
lan743x0: flags=8a63<UP, BROADCAST, RUNNING, ALLMULTI, SIMPLEX, MULTICAST> metric 0 mtu 1500
options=800103<RXCSUM, TXCSUM, TS04>
ether 00:80:0f:74:31:00
inet 192.150.0.100 netmask 0xffffffff broadcast 192.150.0.255
inet6 fe80::280:fff:fe74:3100%lan743x0 prefixlen 64 scopeid 0x6
media: Ethernet 1000BASE-T1 <full-duplex, master>
status: active
nd6 options=21<PERFORMNUD, AUTO_LINKLOCAL>
```

5 Driver sysctl parameters

The LAN743x driver/device statistics can be obtained using below command.

```
# sysctl dev.lan743x.0
```

The LAN887x driver parameters can be displayed using,

```
# sysctl dev.lan887x.0
```

6 LAN887x Phy diagnostics

The LAN887x driver supports SQR and cable diagnostics through *sysctl* commands.

6.1 Signal Quality Indicator (SQI)

The `sysctl dev.lan887x.0.sqi` command displays the SQI value. The value 7 means good link.

```
# sysctl dev.lan887x.0.sqi
dev.lan887x.0.sqi: 7
#
```

6.2 Cable Diagnostics

The cable diagnostic can be started using `sysctl dev.lan887x.0.cd_test` command. The diagnostic status can be obtained using `sysctl dev.lan887x.0.cd_report` command. The status and distance will be displayed.

When link is up:

```
# sysctl dev.lan887x.0.cd_test
dev.lan887x.0.cd_test: started
# sysctl dev.lan887x.0.cd_report
dev.lan887x.0.cd_report: status(ok) distance(0m 0cm)
#
```

When cable break or removed:

```
# sysctl dev.lan887x.0.cd_test
dev.lan887x.0.cd_test: started
# sysctl dev.lan887x.0.cd_report
dev.lan887x.0.cd_report: status(open) distance(1m 472cm)
#
```

7 Driver module load parameters

The LAN743x and LAN887x drivers supports module load parameters via config file. Using below parameters the driver settings can be changed at the time of driver mount.

```
dev.lan887x.0.strap_override="1"
dev.lan887x.0.speed_1000="1"
dev.lan887x.0.mode_master="1"
dev.lan887x.0.aneg_enable="1"
dev.lan743x.0.phy_addr="21"
```

The Phy address can be changed using the LAN743x driver parameter `dev.lan743x.0.phy_addr`. Please make sure to match your board design. The phy parameters can be changed using the parameters `dev.lan887x`. The default values are listed as above.

The parameters need to be in a configuration file that can be used with io-sock command.

Below example shows how to start the alternate network stack.

1. Create config file

```
# echo "dev.lan887x.0.strap_override=\"1\" \ndev.lan887x.0.speed_1000=\"1\" \ndev.lan887x.0.mode_master=\"1\" \ndev.lan887x.0.aneg_enable=\"1\" \ndev.lan743x.0.phy_addr=\"21\"\" >> /tmp/override.conf
```

2. Start alternate network stack and mount the drivers

```
# io-sock -m mods-pci.so -m mods-phy.so -o config=/tmp/override.conf -o prefix=/alt
# mount -T io-sock -o prefix=/alt /tmp/devs-lan887x.so
# mount -T io-sock -o prefix=/alt /tmp/devs-lan743x.so
```

3. Configure IP address

```
# SOCK=/alt ifconfig lan743x0 192.168.2.80/24 up
```

Using the alternate stack requires all the network commands to be invoked with SOCK variable.

8 Compile time configurations

If compile time configurations are preferred choice while building the driver rather than using module parameters, the source code can be modified to provide the parameters.

1. LAN743x driver

Set the Phy address of the phy connected to LAN7431 in the *lan743x_reg_defs.h* file.

```
#define LAN743X_LAN7431_PHY_ADDRESS (1)
```

2. LAN887x driver

The *load_cfg_sysctl_overrides()* function in the *lan887x.c* file can be modified to enable the required configuration.

```
phy->strap_override = true; /* Use software settings, below
speed/master/slave/duplex settings apply */
```

```
phy->cfg.speed_1000 = true; /* speed 1000 when true, 100 when false */
```

```
phy->cfg.master = true; /* master when true, slave when false */
```

```
phy->cfg.aneg = true; /* autoneg enabled when true */
```

9 X86_64 BSP Inclusion

This section describes how to include the drivers in the x86_64 board support package. The similar procedure can be followed for other platform BSPs.

The driver .so files must be copied to the architecture related /lib/dll folder (ex: BSP_project/install/x86_64/lib/dll). Also, respective build file (ex: for uefi build, modify BSP_project/images/generic-uefi/ x86_64-generic-uefi.build) must be modified to generate config file with the module parameters. The '*startup-script={*' must be modified to load the driver as in the example described in the section 7. Use of '-o prefix=' to io-sock is optional and only required if second network stack need to be instantiated.